GRK Test 1 · every task type + answer key

lectures 1–6 · Szymański · Fri 13 Nov 2026 · 15 pts

34 task types, 73 problems. Each type has the **official / past problem** (from the 2025 "Studying Graphics" guide, the exam papers 2020–2023 and the slides) and **practice problems with new numbers**. The answer key starts on a new page, so you can tear it off or fold it back.

How to use: do the problem on squared paper first (formula → steps table → drawing → boxed answer), no calculator, then check with the key. Mark every problem you got wrong and redo it tomorrow. The answers follow the lecturer's conventions (Bresenham $p \geq 0 \rightarrow$ up; circle $p \geq 0 \rightarrow y-1$; codes TBRL; clip order left, top, right, bottom; M = last·…·first; frame $+\theta$ = point $-\theta$). Where the 2025 guide is wrong, the key says so in a red box.

Out of scope for Test 1: part 2 questions 6–25 of the 2025 guide (3D representations, CSG, scene graphs, shadows, lights, quadtrees/octrees, 3D transforms, LookAt, perspective, back-face, z-buffer, frustum, shading) belong to lectures 7+ → Test 2 / exam.

#	task type	problems (source)
L1 · Displays, pipeline, vectors		
1	Data rate of a display	1a SG1 Q1 · 1b practice
2	Time to display one pixel	2a SG1 Q2 · 2b practice
3	Memory of a vector image	3a SG1 Q3 · 3b practice
4	Stages of the graphics pipeline	4a SG1 Q4
5	Fragment shader input and output	5a SG1 Q5 · 5b practice
6	Vectors: components, sum, cross product, matrix × vector	6a SG1 Q28 · 6b SG1 Q29 · 6c practice
L2 · Image representation		
7	Memory of a bitmap	7a SG1 Q6 · 7b practice
8	Linear resolution (ppi) of a display	8a SG1 Q7 · 8b practice · 8c practice
9	Maximum useful resolution (human eye)	9a SG1 Q8 · 9b practice
10	Pseudo-code: decrease resolution 2:1	10a SG1 Q9
11	Resampling: intensity of a new pixel (bilinear)	11a SG1 Q16 · 11b practice
12	Sketch the image resampled to lower ppi	12a SG1 Q27
L3 · Colour		
13	How many colours for b bits per channel	13a SG1 Q10 · 13b practice
14	Reduce colour/grey depth (8 → n bits)	14a SG1 Q11 · 14b EXAM 1/23 #1 · 14c practice
15	RGB → CMY	15a SG1 Q12 · 15b practice
16	RGB → greyscale	16a SG1 Q13 · 16b practice
17	Mix two RGB colours	17a SG1 Q15 · 17b practice
18	Chromaticity diagram: what can be mixed	18a SG1 Q14
19	Describe the RGB model	19a SG1 Q17
20	Mark colours in the HSV cone	20a EXAM 1/23 #1c · 20b practice
L4 · Lines and antialiasing		
21	DDA: steps and pixels	21a SG1 Q18 · 21b practice
22	Why slope ≤ 45° in basic DDA	22a SG1 Q19
23	Bresenham: steps and pixels	23a SG1 Q20 · 23b lecture slides · 23c practice
24	Antialiased version of a line	24a SG1 Q25 · 24b practice
L5 · Circles, fills, triangles		
25	Midpoint circle	25a SG1 Q21 · 25b practice
26	Scan-line fill	26a SG1 Q22 · 26b practice
27	Seed (starting-point) fill: order of pixels	27a SG1 Q23 · 27b practice
28	Barycentric: inside test and colour	28a SG1 Q24 + Q26 · 28b EXAM 1/22 #3 · 28c practice
L6 · 2D transforms and frames		
29	Compose 2D transformations and apply	29a SG2 Q1 · 29b Exam 1 2020 T4 · 29c EXAM 1/23 #3 · 29d practice · 29e practice
30	Change of coordinate frame	30a SG2 Q2 · 30b practice · 30c practice
L6 · Clipping		
31	Cohen–Sutherland line clipping	31a SG2 Q3 · 31b Exam 1 2020 groups 1–4 · 31c practice
32	Sutherland–Hodgman polygon clipping	32a SG2 Q4 · 32b practice
L6 · Bézier curves		
33	Bézier curve construction (+ smooth joining)	33a SG2 Q5 · 33b EXAM 1/22 #1 · 33c practice · 33d practice
Theory · true / false with a reason		
34	Short theory / true-false	34a lecture tests, older papers

Part 1 · Problems

answers in Part 2 · source = past/official · practice (new numbers)

L1 · Displays, pipeline, vectors

1. Data rate of a display

1a Studying Graphics 2025, part 1 Q1 What is the data rate (bytes/s) needed to display a 4K image at 60 Hz (24-bit colour)?

1b practice (new numbers) Full HD 1920×1080, 24-bit colour, 60 Hz. Data rate?

2. Time to display one pixel

2a Studying Graphics 2025, part 1 Q2 1920×1080 display, 80 Hz. Time needed to display a single pixel?

2b practice (new numbers) 4K UHD 3840×2160 at 60 Hz. Time per pixel?

3. Memory of a vector image

3a Studying Graphics 2025, part 1 Q3 Memory for 100 circles and 100 triangles in a simple vector representation (4-byte floats)?

3b practice (new numbers) 50 circles, 20 rectangles and 30 line segments, 4-byte floats.

4. Stages of the graphics pipeline

4a Studying Graphics 2025, part 1 Q4 Describe the stages of the graphics pipeline.

5. Fragment shader input and output

5a Studying Graphics 2025, part 1 Q5 What is the input and output data of the fragment shader?

5b practice (new numbers) And the vertex shader?

6. Vectors: components, sum, cross product, matrix × vector

6a Studying Graphics 2025, part 1 Q28 A(1,1,1), B(3,1,1), C(1,2,1) in a right-handed system. Draw vectors AB and AC and write their components.

6b Studying Graphics 2025, part 1 Q29 X = [–2,0,0], Y = [0,1,0] drawn from P(–1,1,0). Sum X + Y, draw it.

6c practice (new numbers) a = [2,0,1], b = [0,3,0]. Compute a + b, a·b, a × b.

L2 · Image representation

7. Memory of a bitmap

7a Studying Graphics 2025, part 1 Q6 Memory for an image 1024 × 768 × 24 in MB.

7b practice (new numbers) 1920 × 1080 × 16 bit in MB (and MiB).

8. Linear resolution (ppi) of a display

8a Studying Graphics 2025, part 1 Q7 42″ (diagonal) display, 1920 × 1080, 16:9. Linear resolution?

8b practice (new numbers) 24″ display, 1920 × 1200, 16:10.

8c practice (new numbers) 27″ display, 2560 × 1440, 16:9.

9. Maximum useful resolution (human eye)

9a Studying Graphics 2025, part 1 Q8 Max horizontal resolution the eye can use: display width 30 cm, viewing distance 50 cm.

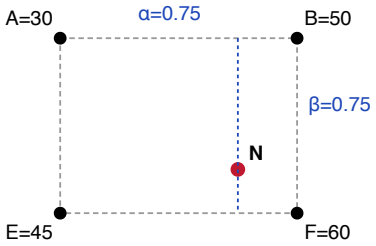
9b practice (new numbers) Width 50 cm, distance 2 m.

10. Pseudo-code: decrease resolution 2:1

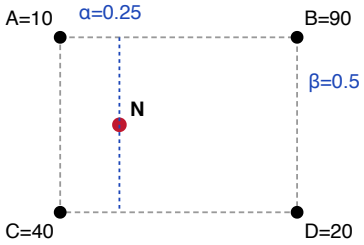
10a Studying Graphics 2025, part 1 Q9 Pseudo-code to decrease / increase resolution 2:1. Image: Color[][] image = new Color[100][100].

11. Resampling: intensity of a new pixel (bilinear)

11a Studying Graphics 2025, part 1 Q16 Resolution changed. A = 30, B = 50 (top row), E = 45, F = 60 (bottom row). New pixel N lies ¾ of the way from A to B and ¾ of the way down. Intensity of N?

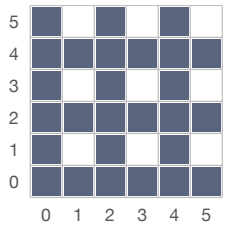


11b practice (new numbers) A = 10, B = 90 (top), C = 40, D = 20 (bottom); α = 0.25 across, β = 0.5 down.



12. Sketch the image resampled to lower ppi

12a Studying Graphics 2025, part 1 Q27 6×6 image sampled at 10 ppi (below). Sketch it at 5 ppi. Is it a good representation? Why?



L3 · Colour

13. How many colours for b bits per channel

13a Studying Graphics 2025, part 1 Q10 How many distinct colours with 3 bits per RGB channel?

13b practice (new numbers) 5 bits per channel? And 16-bit "high colour" 5-6-5?

14. Reduce colour/grey depth (8 → n bits)

14a Studying Graphics 2025, part 1 Q11 Grey levels 143, 240, 128 at 8 bits. Give them at 4 bits.

14b EXAM 1/23 #1 Circle [242,254,4], triangle [240,0,241], rectangle [247,13,253] (8 bit/ch). Reduce to 3 bits/ch, count the colours, mark them in the HSV cone.

14c practice (new numbers) Grey 200, 100, 37, 255 at 8 bits → 3 bits.

15. RGB → CMY

15a Studying Graphics 2025, part 1 Q12 RGB → CMY: [255,135,0], [0,0,0], [255,255,255].

15b practice (new numbers) RGB [200,50,100] → CMY. Then CMY [0.2, 0.5, 1] → RGB (8 bit).

16. RGB → greyscale

16a Studying Graphics 2025, part 1 Q13 RGB → grey: [128,0,128], [255,255,0].

16b practice (new numbers) RGB [100,200,50] → grey.

17. Mix two RGB colours

17a Studying Graphics 2025, part 1 Q15 Mix equal parts: A = [255,255,0], B = [0,255,255].

17b practice (new numbers) 30 % red [255,0,0] + 70 % blue [0,0,255].

18. Chromaticity diagram: what can be mixed

18a Studying Graphics 2025, part 1 Q14 Colours 1 and 2 are marked on the xy chromaticity diagram. Which colours can be obtained by mixing them?

19. Describe the RGB model

19a Studying Graphics 2025, part 1 Q17 Describe the RGB model: components, coordinate system (draw axes), mark white, black, yellow, write them numerically, applications.

20. Mark colours in the HSV cone

20a EXAM 1/23 #1c Mark [7,7,0] and [7,0,7] (3-bit) in the HSV cone.

20b practice (new numbers) Mark [0,128,255] (8 bit) and [3,3,3] (3 bit) in the HSV cone.

L4 · Lines and antialiasing

21. DDA: steps and pixels

21a Studying Graphics 2025, part 1 Q18 DDA steps and pixels from (0,0) to (5,3).

21b practice (new numbers) DDA from (1,1) to (7,4).

22. Why slope ≤ 45° in basic DDA

22a Studying Graphics 2025, part 1 Q19 Why is the slope assumed ≤ 45° in basic DDA? Show an example.

23. Bresenham: steps and pixels

23a Studying Graphics 2025, part 1 Q20 Bresenham steps and pixels from (0,0) to (5,2).

23b lecture slides Bresenham from (2,2) to (8,5) (slide example).

23c practice (new numbers) Bresenham from (8,5) to (1,1).

24. Antialiased version of a line

24a Studying Graphics 2025, part 1 Q25 How would an antialiased version of the Q20 line (0,0)→(5,2) look? Principle?

24b practice (new numbers) Antialiased line (0,0) → (6,2).

L5 · Circles, fills, triangles

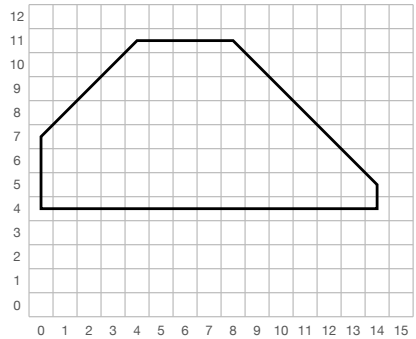
25. Midpoint circle

25a Studying Graphics 2025, part 1 Q21 Midpoint circle: centre (0,0), radius 10. Steps and pixels.

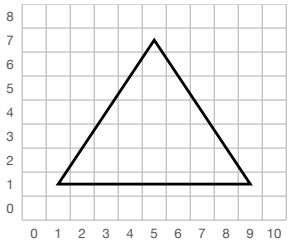
25b practice (new numbers) Midpoint circle, r = 8.

26. Scan-line fill

26a Studying Graphics 2025, part 1 Q22 Scan-line fill of the "hill" shape (0,4) (14,4) (14,5) (8,11) (4,11) (0,7). Describe the steps.

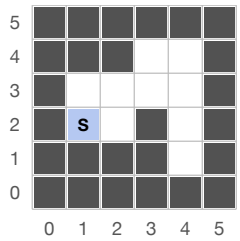


26b practice (new numbers) Scan-line fill of the triangle (1,1), (9,1), (5,7).

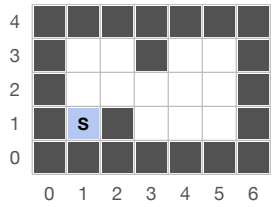


27. Seed (starting-point) fill: order of pixels

27a Studying Graphics 2025, part 1 Q23 Starting-point (seed) fill: describe the steps and mark the order in which pixels are drawn. Seed (1,2).

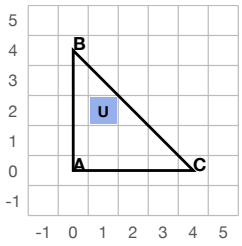


27b practice (new numbers) Seed fill from (1,1), order left, right, up, down.



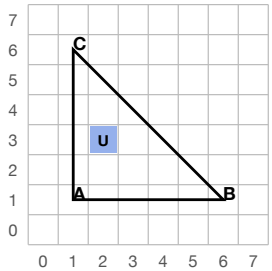
28. Barycentric: inside test and colour

28a Studying Graphics 2025, part 1 Q24 + Q26 Triangle A(0,0), B(0,4), C(4,0); pixel U(1,2). Inside? Grey levels at the vertices: lecture version A 255, B 0, C 128; guide version A 10, B 25, C 128.



28b EXAM 1/22 #3 Texture: (0,0)→uv(0,0), (7,0)→uv(1,0), (7,7)→uv(1,1). Pixel (4,3): which texel?

28c practice (new numbers) A(1,1), B(6,1), C(1,6). Is (2,3) inside? Grey at A, B, C = 0, 100, 200. Is (5,4) inside?



L6 · 2D transforms and frames

29. Compose 2D transformations and apply

29a Studying Graphics 2025, part 2 Q1 Triangle (1,1), (3,1), (1,3): translate by [1,−2], then scale sx = 0.5, sy = 1. Matrix first, then the points.

29b Exam 1 2020 T4 Translate, then rotate 45° about the origin. Group 1: (1,1), (3,1), (1,3), T[1,–2] · group 2: (1,2), (3,2), (1,4), T[–1,2] · group 3: (2,1), (4,1), (2,3), T[–1,–2] · group 4: (2,2), (4,2), (2,4), T[1,2].

29c EXAM 1/23 #3 (–5,0), (–1,0), (–1,4): scale sx = 1.5, sy = 0.5 about the point (–3,2).

29d practice (new numbers) (1,0), (3,0), (1,2): rotate 90° about the origin, then translate by [2,1].

29e practice (new numbers) (1,1), (3,1), (1,3): scale ×2 about (1,1).

30. Change of coordinate frame

30a Studying Graphics 2025, part 2 Q2 P = (2,2) in OXY. Frame EUV is rotated 90° counter-clockwise relative to OXY. (u,v) of P?

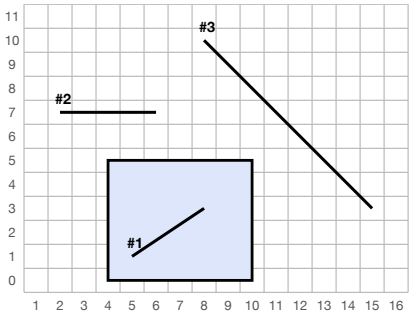
30b practice (new numbers) P = (3,1); frame rotated 90° clockwise (same origin).

30c practice (new numbers) P = (4,3); frame with origin E(1,1), rotated 90° counter-clockwise.

L6 · Clipping

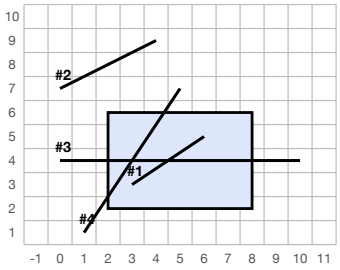
31. Cohen–Sutherland line clipping

31a Studying Graphics 2025, part 2 Q3 Window (4,0)–(10,5). Segments (5,1)–(8,3), (2,7)–(6,7), (8,10)–(15,3). Cohen–Sutherland steps.



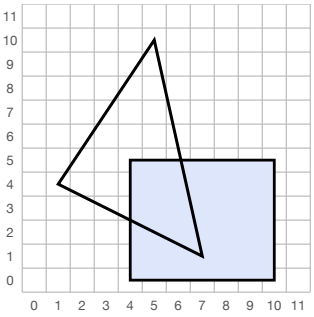
31b Exam 1 2020 groups 1–4 Same window (4,0)–(10,5). Group 1: (3,1)–(13,3), (2,8)–(6,8), (7,10)–(15,3) · group 2: (3,2)–(13,4), (2,6)–(6,6), (7,11)–(15,4) · group 3: (2,3)–(12,4), (1,6)–(7,6), (8,11)–(14,4) · group 4: (3,2)–(11,4), (3,7)–(7,7), (9,11)–(15,5).

31c practice (new numbers) Window (2,2)–(8,6). Segments #1 (3,3)–(6,5), #2 (0,7)–(4,9), #3 (0,4)–(10,4), #4 (1,1)–(5,7).

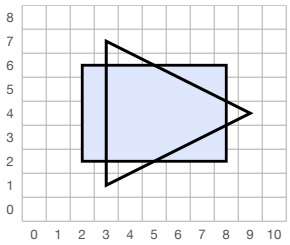


32. Sutherland–Hodgman polygon clipping

32a Studying Graphics 2025, part 2 Q4 Triangle (7,1), (1,4), (5,10); window (4,0)–(10,5). Sutherland–Hodgman steps.



32b practice (new numbers) Triangle (3,1), (9,4), (3,7); window (2,2)–(8,6).



L6 · Bézier curves

33. Bézier curve construction (+ smooth joining)

33a Studying Graphics 2025, part 2 Q5 Quadratic Bézier: endpoints (1,1) and (10,1), control point (4,8). Explain the construction, show it for t = 0.25, 0.5, 0.75, draw the curve.

33b EXAM 1/22 #1 Cubic Bézier $P_0(1,1)$, $P_1(3,-7)$, $P_2(8,-7)$, $P_3(10,1)$: points at t = 0.25, 0.5, 0.75, and the first control point of a smoothly joined next curve.

33c practice (new numbers) Quadratic (0,0), (2,6), (8,0).

33d practice (new numbers) Cubic (0,0), (0,4), (6,4), (6,0).

Theory · true / false with a reason

34. Short theory / true-false

34a lecture tests, older papers True or false? Give a one-line reason each.

1. A pixel has a size.
2. Antialiasing removes aliasing completely.
3. CMY is an additive model.
4. Bresenham uses only integer additions.
5. Two rotations about the same point commute.
6. Cohen–Sutherland: AND of the codes = 0000 means the segment is visible.
7. A Bézier curve passes through all its control points.
8. Refresh time of a raster display depends on the picture.
9. An RGB monitor can show every visible colour.
10. The fragment shader outputs transformed vertex positions.
11. Homogeneous coordinates let translation be a matrix multiplication.
12. Spectral (monochromatic) colours lie on the straight edge of the CIE diagram.
13. Lowering the resolution lowers the memory needed.
14. Sutherland–Hodgman clips against all window edges at once.

Part 2 · Answer key

worked like a model answer · numbers from the tested solver

L1 · Displays, pipeline, vectors

1. Data rate of a display

1a Studying Graphics 2025, part 1 Q1

1. pixels per frame = 4096 × 2160 = 8,847,360
2. bytes per pixel = 24 bits ÷ 8 = 3 B
3. data rate = pixels × bytes/px × refresh = 8,847,360 × 3 × 60 Hz = 1,592,524,800 B/s
4. in bits/s: × 8 = 1.274·10^10 b/s
5. Lecture slide 26 takes 4K = 4096×2160 and rounds up to 10^7 pixel-clock ticks (blanking included): 10^7 × 3 B × 60 = $1.8 \cdot 10^9$ B/s ≈ "1,7 GB/s" (binary GiB). Write the exact pixel count and mention that rounding.

Answer: slide answer $1.8 \cdot 10^9$ B/s (4096×2160 ≈ 10^7 ticks incl. blanking); exact pixels $1.59 \cdot 10^9$ B/s

Write both: the slide rounds 4096·2160 = 8 847 360 up to 10^7 ticks.

1b practice (new numbers)

1. pixels per frame = 1920 × 1080 = 2,073,600
2. bytes per pixel = 24 bits ÷ 8 = 3 B
3. data rate = pixels × bytes/px × refresh = 2,073,600 × 3 × 60 Hz = 373,248,000 B/s
4. in bits/s: × 8 = 2.986·10^9 b/s

Answer: 373,248,000 B/s ≈ 3.732·10^8 B/s ≈ 0.373 GB/s (0.348 GiB/s)

By hand: 1920·1080 = 2 073 600 ≈ $2.07 \cdot 10^6$; ·3 = $6.22 \cdot 10^6$; ·60 = $3.73 \cdot 10^8$ B/s.

2. Time to display one pixel

2a Studying Graphics 2025, part 1 Q2

1. pixels per second = 1920 × 1080 × 80 = 165,888,000
2. time per pixel = 1 ÷ 165,888,000 s = 6.028·10^-9 s = 6.0282 ns
3. (per frame: 1/80 s = 12.5 ms; per line: 11.5741 µs)

Answer: ≈ 6.03 ns

> The 2025 guide says "6 µs" — wrong by 1000×. 12.5 ms / 2 073 600 = $6.03 \cdot 10^{-9}$ s.

2b practice (new numbers)

1. pixels per second = 3840 × 2160 × 60 = 497,664,000
2. time per pixel = 1 ÷ 497,664,000 s = 2.009·10^-9 s = 2.0094 ns
3. (per frame: 1/60 s = 16.6667 ms; per line: 7.716 µs)

Answer: t = 1/(3840·2160·60) = 2.009·10^-9 s ≈ 2.009 ns

3. Memory of a vector image

3a Studying Graphics 2025, part 1 Q3

shape	count	numbers each	bytes each	subtotal
circle	100	3	3·4 = 12 B	1,200 B
triangle	100	6	6·4 = 24 B	2,400 B

Answer: 3,600 B

3b practice (new numbers)

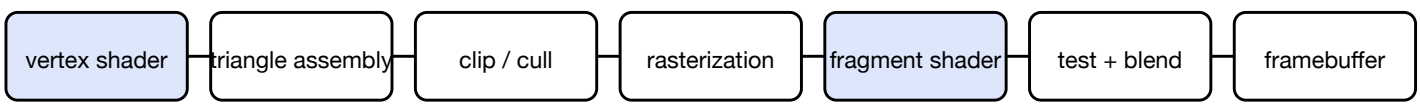
shape	count	numbers each	bytes each	subtotal
circle	50	3	3·4 = 12 B	600 B
rectangle	20	8	8·4 = 32 B	640 B
line	30	4	4·4 = 16 B	480 B

Answer: 1,720 B

Circle x, y, r = 3 numbers; rectangle 4 corners = 8; line 2 endpoints = 4.

4. Stages of the graphics pipeline

4a Studying Graphics 2025, part 1 Q4



blue = programmable · vertices → primitives → fragments → pixels

1. **Application (CPU)**: scene, objects, culling, draw calls.
2. **Vertex shader***: transforms each vertex (model → view → projection), passes data on.

3. **Primitive (triangle) assembly**, then **clipping** and back-face **culling**.
4. **Rasterization**: triangle → fragments (candidate pixels) with interpolated values.
5. **Fragment shader***: colour of each fragment (texture, lighting).
6. **Tests + blending**: depth test, alpha blending → **frame buffer** → display.

* programmable stages.

5. Fragment shader input and output

5a Studying Graphics 2025, part 1 Q5

Input: one fragment with values interpolated by the rasterizer (colour, texture coordinates UV, normal, position/depth) + uniforms (textures, lights, matrices). **Output**: the fragment colour RGBA (optionally depth). Runs once per fragment → the most expensive stage.

5b practice (new numbers)

Input: vertex attributes (position, colour, normal, UV) + matrices. **Output**: transformed (clip-space) position + values to be interpolated for the fragments.

6. Vectors: components, sum, cross product, matrix × vector

6a Studying Graphics 2025, part 1 Q28

AB = B − A = **[2,0,0]** (along +x), AC = C − A = **[0,1,0]** (along +y). Draw both starting at A; right-handed: x right, y up, z toward you.

6b Studying Graphics 2025, part 1 Q29

X + Y = **[−2,1,0]**; from P its tip is at (−3,2,0) (tip-to-tail). a × b = [a_y b_z − a_z b_y, a_z b_x − a_x b_z, a_x b_y − a_y b_x] = [0·0 − 0·1, 0·0 − -2·0, -2·1 − 0·0] = [0, 0, -2]

6c practice (new numbers)

1. a + b = [2, 0, 1] + [0, 3, 0] = [2, 3, 1] (draw b starting at the tip of a)
2. b − a = [-2, 3, -1]
3. a · b = 2·0 + 0·3 + 1·0 = 0 → angle = 90°
4. a × b = [a_y b_z − a_z b_y, a_z b_x − a_x b_z, a_x b_y − a_y b_x] = [0·0 − 1·3, 1·0 − 2·0, 2·3 − 0·0] = [-3, 0, 6]
5. |a| = 2.2361, |b| = 3, |a × b| = 6.7082 (= area of the parallelogram)
6. Right-hand rule: fingers along a, curl toward b, thumb = a × b. x × y = z, y × z = x, z × x = y; b × a = −(a × b).

Answer: a + b = [2, 3, 1] · a × b = [-3, 0, 6] · a·b = 0

a·b = 0 → perpendicular. Check a × b ⊥ a: [−3,0,6]·[2,0,1] = −6 + 6 = 0 ✓.

L2 · Image representation

7. Memory of a bitmap

7a Studying Graphics 2025, part 1 Q6

1. bits = 1024 × 768 × 24 = 18,874,368 bits
2. bytes = 18,874,368 ÷ 8 = 2,359,296 B
3. MB (10^6) = 2.3593 MB; MiB (2^{20} = 1,048,576) = 2.25 MiB; KB = 2359.3 kB / 2304 KiB
4. Write which megabyte you mean. Both are accepted if you say it.

Answer: 2,359,296 B = 2.3593 MB (2.25 MiB)

7b practice (new numbers)

1. bits = 1920 × 1080 × 16 = 33,177,600 bits
2. bytes = 33,177,600 ÷ 8 = 4,147,200 B
3. MB (10^6) = 4.1472 MB; MiB (2^{20} = 1,048,576) = 3.9551 MiB; KB = 4147.2 kB / 4050 KiB
4. Write which megabyte you mean. Both are accepted if you say it.

Answer: 4,147,200 B = 4.1472 MB (3.9551 MiB)

8. Linear resolution (ppi) of a display

8a Studying Graphics 2025, part 1 Q7

1. aspect 16:9 → diagonal units = $\sqrt{(16^2 + 9^2)} = \sqrt{337} = 18.3576$
2. width = 42 · 16/18.3576 = 36.6062″, height = 42 · 9/18.3576 = 20.591″
3. linear resolution = 1920 px ÷ 36.6062″ = 52.45 ppi (check: 1080 ÷ 20.591 = 52.45 ppi)
4. pixel size = 25.4 mm ÷ 52.45 = 0.484 mm

Answer: 52.45 ppi (width 36.606″, height 20.591″)

8b practice (new numbers)

1. aspect 16:10 → diagonal units = $\sqrt{(16^2 + 10^2)} = \sqrt{356} = 18.868$
2. width = 24 · 16/18.868 = 20.352″, height = 24 · 10/18.868 = 12.72″
3. linear resolution = 1920 px ÷ 20.352″ = 94.34 ppi (check: 1200 ÷ 12.72 = 94.34 ppi)
4. pixel size = 25.4 mm ÷ 94.34 = 0.269 mm

Answer: 94.34 ppi (width 20.352″, height 12.72″)

8c practice (new numbers)

1. aspect 16:9 → diagonal units = $\sqrt{(16^2 + 9^2)} = \sqrt{337} = 18.3576$

2. width = 27 · 16/18.3576 = 23.5325″, height = 27 · 9/18.3576 = 13.2371″
3. linear resolution = 2560 px ÷ 23.5325″ = 108.786 ppi (check: 1440 ÷ 13.2371 = 108.786 ppi)
4. pixel size = 25.4 mm ÷ 108.79 = 0.233 mm

Answer: 108.79 ppi (width 23.533″, height 13.237″)

By hand: width = 27 · 0.8716 = 23.53″; 2560 / 23.53 ≈ 108.8.

9. Maximum useful resolution (human eye)

9a Studying Graphics 2025, part 1 Q8

1. The eye separates details down to about 1 arc-minute = 1/60°.
2. smallest detail at distance D: x = D · tan(1/60°) = 50 · 0.0002909 = 0.01454 (same unit as D)
3. max pixels across the width: n = W / x = 30 / 0.01454 = 2062.6
4. Slide 20 rounds tan 1′ ≈ 0.00029: x = 50·0.00029 = 0.0145 → n = 2069 (the lecturer's number for 30 cm @ 50 cm is ≈ 2069). Show the formula; either value is fine.
5. More pixels than that are wasted: the eye cannot resolve them at this distance.

Answer: ≈ 2063 px (slide rounding tan 1′ = 0.00029 → 2069)

9b practice (new numbers)

1. The eye separates details down to about 1 arc-minute = 1/60°.
2. smallest detail at distance D: x = D · tan(1/60°) = 200 · 0.0002909 = 0.05818 (same unit as D)
3. max pixels across the width: n = W / x = 50 / 0.05818 = 859.4
4. Slide 20 rounds tan 1′ ≈ 0.00029: x = 200·0.00029 = 0.058 → n = 862.1 (the lecturer's number for 30 cm @ 50 cm is ≈ 2069). Show the formula; either value is fine.
5. More pixels than that are wasted: the eye cannot resolve them at this distance.

Answer: ≈ 859 pixels across (smallest visible detail 0.5818 mm) (slide rounding: 50 / (200·0.00029) = 862)

10. Pseudo-code: decrease resolution 2:1

10a Studying Graphics 2025, part 1 Q9

```
Color[][] half(Color[][] img) {
  Color[][] out = new Color[50][50];
  for (int y = 0; y < 50; y++)
    for (int x = 0; x < 50; x++) {
      Color a = img[2*y][2*x],   b = img[2*y][2*x+1];
      Color c = img[2*y+1][2*x], d = img[2*y+1][2*x+1];
      out[y][x] = new Color((a.r+b.r+c.r+d.r)/4,
                            (a.g+b.g+c.g+d.g)/4,
                            (a.b+b.b+c.b+d.b)/4);
    }
  return out;
}
Color[][] twice(Color[][] img) {          // 1:2, nearest neighbour
  Color[][] big = new Color[200][200];
  for (int y = 0; y < 200; y++)
    for (int x = 0; x < 200; x++) big[y][x] = img[y/2][x/2];
  return big;
}
```

Taking only img[2y][2x] is also accepted if you say it loses information (more aliasing).

11. Resampling: intensity of a new pixel (bilinear)

11a Studying Graphics 2025, part 1 Q16

1. Layout: A —— B on the first row, C —— D on the second; α = horizontal fraction from A toward B, β = vertical fraction from row AB toward row CD.
2. P1 = α·B + (1−α)·A = 0.75·50 + 0.25·30 = 45
3. P2 = α·D + (1−α)·C = 0.75·60 + 0.25·45 = 56.25
4. N = β·P2 + (1−β)·P1 = 0.75·56.25 + 0.25·45 = 53.4375
5. Check: N must lie between the smallest and largest of A, B, C, D.

Answer: N = 53.4375 (≈ 53)

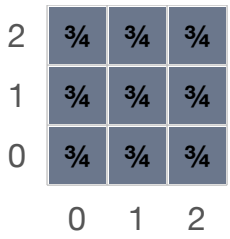
11b practice (new numbers)

1. Layout: A —— B on the first row, C —— D on the second; α = horizontal fraction from A toward B, β = vertical fraction from row AB toward row CD.
2. P1 = α·B + (1−α)·A = 0.25·90 + 0.75·10 = 30
3. P2 = α·D + (1−α)·C = 0.25·20 + 0.75·40 = 35
4. N = β·P2 + (1−β)·P1 = 0.5·35 + 0.5·30 = 32.5
5. Check: N must lie between the smallest and largest of A, B, C, D.

Answer: N = 32.5 (≈ 33)

12. Sketch the image resampled to lower ppi

12a Studying Graphics 2025, part 1 Q27



5 ppi = each new pixel covers a **2×2 block**. Every block has 3 dark + 1 white pixel → averaging gives the **same 75 % grey** in all 9 new pixels (3×3).

With point sampling (take every 2nd pixel) all new pixels are dark → the white dots vanish.

Not a good representation: the pattern repeats every 2 old pixels = one new pixel → it is finer than the new sampling can show (undersampling → aliasing); the detail is lost.

L3 · Colour

13. How many colours for b bits per channel

13a Studying Graphics 2025, part 1 Q10

$(2^3)^3 = 8^3 =$ **512** colours (8 levels per channel, 3 channels, 9 bits in total).

> The 2025 guide answers "8" — that is levels per channel, not colours.

13b practice (new numbers)

(2^5)³ = 32³ = 32,768 colours. 5-6-5: $2^5 \cdot 2^6 \cdot 2^5 = 2^{16} =$ **65 536**.

14. Reduce colour/grey depth (8 → n bits)

14a Studying Graphics 2025, part 1 Q11

8-bit value	÷ 2^4	4-bit (floor, main)	4-bit (round v·15/255)	floor shown back as 8-bit
143	143 / 16 = 8.938	8	8	136
240	240 / 16 = 15	15	14	255
128	128 / 16 = 8	8	8	136

Answer: 143 → 8, 240 → 15, 128 → 8

143 and 128 both become 8 → the object merges with the background.

14b EXAM 1/23 #1

object	8-bit	3-bit (÷32, floor)	HSV
circle	[242,254,4]	[7,7,0]	yellow H 60° S 1 V 1 → top rim
triangle	[240,0,241]	[7,0,7]	magenta H 300° S 1 V 1 → top rim
rectangle	[247,13,253]	[7,0,7]	magenta → same point as the triangle

Count: $(2^3)^3 =$ **512**. Triangle and rectangle become identical → they merge.

14c practice (new numbers)

8-bit value	÷ 2^5	3-bit (floor, main)	3-bit (round v·7/255)	floor shown back as 8-bit
200	200 / 32 = 6.25	6	5	219
100	100 / 32 = 3.125	3	3	109
37	37 / 32 = 1.156	1	1	36
255	255 / 32 = 7.969	7	7	255

Answer: 200 → 6, 100 → 3, 37 → 1, 255 → 7

15. RGB → CMY

15a Studying Graphics 2025, part 1 Q12

C = 255 − R, M = 255 − G, Y = 255 − B → **[0,120,255]**, **[255,255,255]**, **[0,0,0]** (normalised: [0, 0.47, 1], [1,1,1], [0,0,0]).

15b practice (new numbers)

CMY = [55, 205, 155]. Back: RGB = 1 − CMY = [0.8, 0.5, 0] → ×255 = **[204, 128, 0]** (127.5 rounds up; orange).

16. RGB → greyscale

16a Studying Graphics 2025, part 1 Q13

Y = 0.299·R + 0.587·G + 0.114·B = 0.299·128 + 0.587·0 + 0.114·128 = 38.272 + 0 + 14.592 = 52.864 → 53

Y = 0.299·R + 0.587·G + 0.114·B = 0.299·255 + 0.587·255 + 0.114·0 = 76.245 + 149.685 + 0 = 225.93 → 226

Average: (128 + 0 + 128)/3 = **85.3**; (255 + 255 + 0)/3 = **170**.

Write both methods; mind the brackets: (R + G + B)/3, not R + G + B/3 (the 2025 guide misses them).

16b practice (new numbers)

1. Y = 0.299·R + 0.587·G + 0.114·B = 0.299·100 + 0.587·200 + 0.114·50 = 29.9 + 117.4 + 5.7 = 153 → 153
2. Green weighs most because the eye is most sensitive to green, blue least.
3. Simple version (lecture also shows it): (R + G + B)/3 = 116.667. Note the brackets: (R+G+B)/3, not R+G+B/3.

Answer: weighted 153 ≈ 153 (plain average 116.67)

17. Mix two RGB colours

17a Studying Graphics 2025, part 1 Q15

1. mix in proportion 0.5 : 0.5 → each channel = 0.5·c1 + 0.5·c2
2. R = 0.5·255 + 0.5·0 = 127.5; G = 255; B = 127.5
3. On the chromaticity diagram the mix lies on the segment between the two colours (closer to the bigger share).
4. (Adding light at full power instead: clamp(c1 + c2) = [255, 255, 255].)

Answer: [128, 255, 128] (exact [127.5, 255, 127.5])

17b practice (new numbers)

1. mix in proportion 0.3 : 0.7 → each channel = 0.3·c1 + 0.7·c2
2. R = 0.3·255 + 0.7·0 = 76.5; G = 0; B = 178.5
3. On the chromaticity diagram the mix lies on the segment between the two colours (closer to the bigger share).
4. (Adding light at full power instead: clamp(c1 + c2) = [255, 0, 255].)

Answer: [77, 0, 179] (exact [76.5, 0, 178.5])

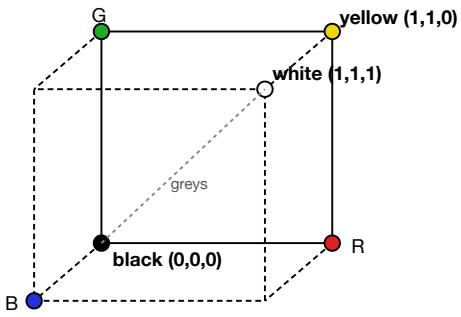
18. Chromaticity diagram: what can be mixed

18a Studying Graphics 2025, part 1 Q14

Exactly the colours on the **straight segment** between 1 and 2 (more of colour 1 → closer to 1). Draw and shade the segment, name 2–3 colours on it. Three primaries → their triangle (the gamut).

19. Describe the RGB model

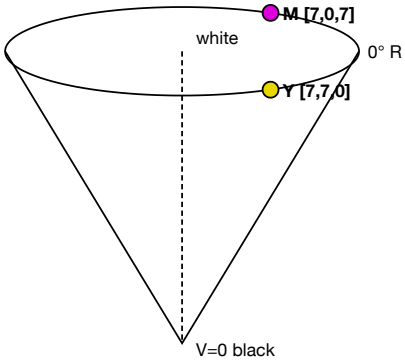
19a Studying Graphics 2025, part 1 Q17



Additive model: the colour is a mix of red, green and blue **light**. Unit cube, axes R, G, B. Black (0,0,0) at the origin; white (1,1,1) opposite; yellow (1,1,0) = R + G. In 8 bits: black [0,0,0], white [255,255,255], yellow [255,255,0]. Greys on the black–white diagonal. Used in emissive devices: monitors, TVs, projectors, cameras, scanners.

20. Mark colours in the HSV cone

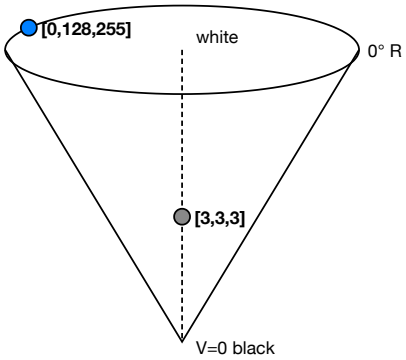
20a EXAM 1/23 #1c



V = max/7 = 1, S = (max − min)/max = 1 → both on the **rim of the top disc**: yellow at H = 60°, magenta at H = 300°.

20b practice (new numbers)

H = 209.9°, S = 1, V = 1 → the rim of the top disc (pure, fully saturated, bright) (between cyan 180° and blue 240°).
H = 0°, S = 0, V = 0.429 → on the central grey axis: a grey, V = 3/7 ≈ 0.43, on the axis below the middle.

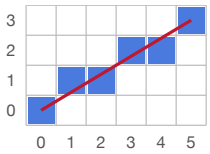


L4 · Lines and antialiasing

21. DDA: steps and pixels

21a Studying Graphics 2025, part 1 Q18

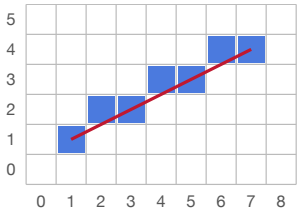
x	y exact	round(y)	pixel
0	0	0	(0,0)
1	0.6	1	(1,1)
2	1.2	1	(2,1)
3	1.8	2	(3,2)
4	2.4	2	(4,2)
5	3	3	(5,3)



21b practice (new numbers)

m = Δy/Δx = 3/6 = 0.5 → |m| ≤ 1, step x by 1, y += m, plot round(y)

x	y exact	round(y)	pixel
1	1	1	(1,1)
2	1.5	2	(2,2)
3	2	2	(3,2)
4	2.5	3	(4,3)
5	3	3	(5,3)
6	3.5	4	(6,4)
7	4	4	(7,4)



22. Why slope ≤ 45° in basic DDA

22a Studying Graphics 2025, part 1 Q19

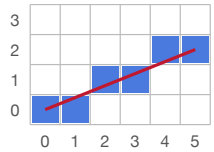
DDA steps x by 1 → one pixel per column. If |m| > 1, y jumps more than 1 per column → **gaps**. Example (0,0) → (2,5): y = 0, 2.5, 5 → only (0,0), (1,3), (2,5); rows 1, 2, 4 empty. Fix: for |m| > 1 step y by 1 and add 1/m to x → (0,0)(0,1)(1,2)(1,3)(2,4)(2,5).

23. Bresenham: steps and pixels

23a Studying Graphics 2025, part 1 Q20

Δx = 5, Δy = 2; a = 2Δy = 4, b = 2Δy − 2Δx = -6, p_0 = 2Δy − Δx = -1

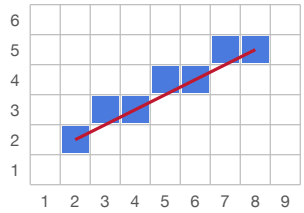
current	p	decision	next	new p
(0,0)	−1	p < 0 → straight	(1,0)	−1 + a = 3
(1,0)	3	p ≥ 0 → step (diagonal)	(2,1)	3 + b = −3
(2,1)	−3	p < 0 → straight	(3,1)	−3 + a = 1
(3,1)	1	p ≥ 0 → step (diagonal)	(4,2)	1 + b = −5
(4,2)	−5	p < 0 → straight	(5,2)	(end)



23b lecture slides

Δx = 6, Δy = 3; a = 2Δy = 6, b = 2Δy − 2Δx = -6, p_0 = 2Δy − Δx = 0

current	p	decision	next	new p
(2,2)	0	p ≥ 0 → step (diagonal)	(3,3)	0 + b = −6
(3,3)	−6	p < 0 → straight	(4,3)	−6 + a = 0
(4,3)	0	p ≥ 0 → step (diagonal)	(5,4)	0 + b = −6
(5,4)	−6	p < 0 → straight	(6,4)	−6 + a = 0
(6,4)	0	p ≥ 0 → step (diagonal)	(7,5)	0 + b = −6
(7,5)	−6	p < 0 → straight	(8,5)	(end)

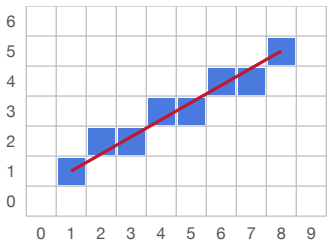


p_0 = 0 → p ≥ 0 → go UP.

23c practice (new numbers)

Start at the smaller x: draw (1,1) → (8,5). Δx = 7, Δy = 4; a = 2Δy = 8, b = 2Δy − 2Δx = -6, p_0 = 2Δy − Δx = 1

current	p	decision	next	new p
(1,1)	1	p ≥ 0 → step (diagonal)	(2,2)	1 + b = −5
(2,2)	−5	p < 0 → straight	(3,2)	−5 + a = 3
(3,2)	3	p ≥ 0 → step (diagonal)	(4,3)	3 + b = −3
(4,3)	−3	p < 0 → straight	(5,3)	−3 + a = 5
(5,3)	5	p ≥ 0 → step (diagonal)	(6,4)	5 + b = −1
(6,4)	−1	p < 0 → straight	(7,4)	−1 + a = 7
(7,4)	7	p ≥ 0 → step (diagonal)	(8,5)	(end)

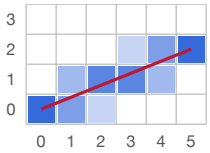


Answer: (1,1) (2,2) (3,2) (4,3) (5,3) (6,4) (7,4) (8,5)

24. Antialiased version of a line

24a Studying Graphics 2025, part 1 Q25

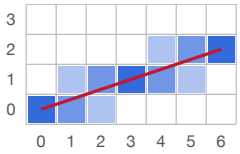
x	true y	lower pixel	upper pixel
0	0	(0,0) 100 %	–
1	0.4	(1,0) 60 %	(1,1) 40 %
2	0.8	(2,0) 20 %	(2,1) 80 %
3	1.2	(3,1) 80 %	(3,2) 20 %
4	1.6	(4,1) 40 %	(4,2) 60 %
5	2	(5,2) 100 %	–



Principle: in each column take the 2 pixels nearest the ideal line; intensity = 1 – distance; the column sums to 100 %. Reduces (does not remove) jaggies.

24b practice (new numbers)

x	true y	lower pixel	upper pixel
0	0	(0,0) 100 %	–
1	0.3333	(1,0) 66.7 %	(1,1) 33.3 %
2	0.6667	(2,0) 33.3 %	(2,1) 66.7 %
3	1	(3,1) 100 %	–
4	1.3333	(4,1) 66.7 %	(4,2) 33.3 %
5	1.6667	(5,1) 33.3 %	(5,2) 66.7 %
6	2	(6,2) 100 %	–



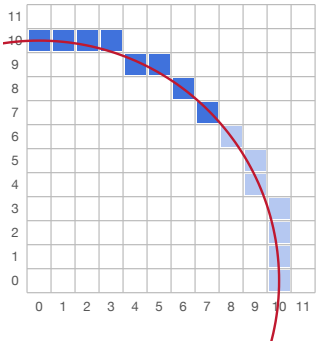
L5 · Circles, fills, triangles

25. Midpoint circle

25a Studying Graphics 2025, part 1 Q21

start (0, r) = (0, 10); p_0 = 5/4 – r = -8.75 (integer version 1 – r = -9 gives the same pixels) p < 0 → (x+1, y), p += $2x_+ + 1$; p ≥ 0 → (x+1, y–1), p += $2x_+ + 1 - 2y_+$ (x_+, y_+ = the NEW x, y)

current	p	decision	next	new p
(0,10)	−8.75	p < 0 → same row	(1,10)	−8.75 + 2·1 + 1 = −5.75
(1,10)	−5.75	p < 0 → same row	(2,10)	−5.75 + 2·2 + 1 = −0.75
(2,10)	−0.75	p < 0 → same row	(3,10)	−0.75 + 2·3 + 1 = 6.25
(3,10)	6.25	p ≥ 0 → one row down	(4,9)	6.25 + 2·4 + 1 − 2·9 = −2.75
(4,9)	−2.75	p < 0 → same row	(5,9)	−2.75 + 2·5 + 1 = 8.25
(5,9)	8.25	p ≥ 0 → one row down	(6,8)	8.25 + 2·6 + 1 − 2·8 = 5.25
(6,8)	5.25	p ≥ 0 → one row down	(7,7)	5.25 + 2·7 + 1 − 2·7 = 6.25

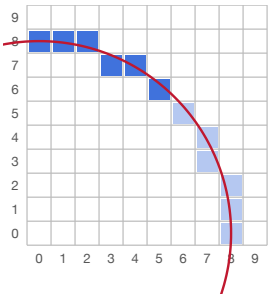


Answer: 1st octant: (0,10) (1,10) (2,10) (3,10) (4,9) (5,9) (6,8) (7,7) → 56 pixels in total after 8-way symmetry

2025 guide: "p ≥ 0 → (x+1, y+1)" — wrong, y goes DOWN: (x+1, y–1).

25b practice (new numbers)

current	p	decision	next	new p
(0,8)	−6.75	p < 0 → same row	(1,8)	−6.75 + 2·1 + 1 = −3.75
(1,8)	−3.75	p < 0 → same row	(2,8)	−3.75 + 2·2 + 1 = 1.25
(2,8)	1.25	p ≥ 0 → one row down	(3,7)	1.25 + 2·3 + 1 − 2·7 = −5.75
(3,7)	−5.75	p < 0 → same row	(4,7)	−5.75 + 2·4 + 1 = 3.25
(4,7)	3.25	p ≥ 0 → one row down	(5,6)	3.25 + 2·5 + 1 − 2·6 = 2.25
(5,6)	2.25	p ≥ 0 → one row down	(6,5)	2.25 + 2·6 + 1 − 2·5 = 5.25



Answer: 1st octant: (0,8) (1,8) (2,8) (3,7) (4,7) (5,6) → 44 pixels in total after 8-way symmetry

26. Scan-line fill

26a Studying Graphics 2025, part 1 Q22

1. For each scan line (row y) find where it crosses the polygon edges: $x = x_1 + (y - y_1)(x_2 - x_1)/(y_2 - y_1)$.
2. Sort the crossings: $x_1 \le x_2 \le x_3 \le \ldots$
3. Fill between pairs: 1st–2nd, 3rd–4th, … (pixels from ceil(x_left) to floor(x_right)).
4. Vertex rule (lecture): edges on opposite sides of the scan line → count the vertex once; both edges on the same side (peak/valley) → count it twice. Horizontal edges are not crossings (they are drawn as contour).

row y	crossings	span incl. outline	interior pixels
11	–	4…8	outline row
10	3, 9	3…9	5
9	2, 10	2…10	7
8	1, 11	1…11	9
7	0, 12	0…12	11
6	0, 13	0…13	12
5	0, 14	0…14	13
4	0, 14	0…14	outline row

Rows 4 and 11 are horizontal outline edges. Rows 5–10: 2 crossings each → 13, 12, 11, 9, 7, 5 interior pixels.

26b practice (new numbers)

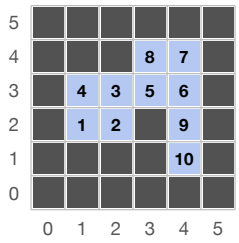
row y	crossings x	fill	pixels
7	5, 5	5…5	1
6	4.33, 5.67	5…5	1
5	3.67, 6.33	4…6	3
4	3, 7	3…7	5
3	2.33, 7.67	3…7	5
2	1.67, 8.33	2…8	7
1	1, 9	1…9	9

Answer: 7 rows filled; total 31 pixels

Row 7: apex = peak → counted twice → 1 pixel. Row 1: horizontal edge = contour.

27. Seed (starting-point) fill: order of pixels

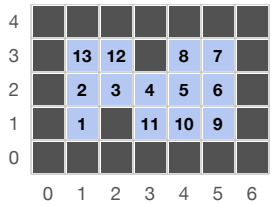
27a Studying Graphics 2025, part 1 Q23



Recursive 4-connected, order left, right, up, down (slide-30 code). 1:(1,2) 2:(2,2) 3:(2,3) 4:(1,3) 5:(3,3) 6:(4,3) 7:(4,4) 8:(3,4) 9:(4,2) 10:(4,1)

27b practice (new numbers)

1:(1,1) 2:(1,2) 3:(2,2) 4:(3,2) 5:(4,2) 6:(5,2) 7:(5,3) 8:(4,3) 9:(5,1) 10:(4,1) 11: (3,1) 12:(2,3) 13:(1,3)



28. Barycentric: inside test and colour

28a Studying Graphics 2025, part 1 Q24 + Q26

1. U = αA + βB + γC with α + β + γ = 1 → three equations:
2. x: 1 = 0α + 0β + 4γ
3. y: 2 = 0α + 4β + 0γ
4. 1 = α + β + γ
5. → α = 0.25, β = 0.5, γ = 0.25 (check: sum = 1)
6. all > 0 → INSIDE
7. value = 0.25·255 + 0.5·0 + 0.25·128 = 95.75 ≈ 96

Answer: inside; lecture values → ≈ 96

Guide values: value = 0.25·10 + 0.5·25 + 0.25·128 = 47 ≈ 47

28b EXAM 1/22 #3

Weights: α = 0.4286, β = 0.1429, γ = 0.4286 (check: sum = 1) all > 0 → INSIDE → (u,v) = (0.571, 0.429) = (4/7, 3/7) → read the texel in that cell of the texture (exam picture: **(20, 20, 100)**).

28c practice (new numbers)

1. U = αA + βB + γC with α + β + γ = 1 → three equations:
2. x: 2 = 1α + 6β + 1γ
3. y: 3 = 1α + 1β + 6γ
4. 1 = α + β + γ
5. → α = 0.4, β = 0.2, γ = 0.4 (check: sum = 1)
6. all > 0 → INSIDE
7. value = 0.4·0 + 0.2·100 + 0.4·200 = 100 ≈ 100

Answer: α = 0.4, β = 0.2, γ = 0.4 → inside · value = 100 ≈ 100

(5,4): α = -0.4, β = 0.8, γ = 0.6 → outside.

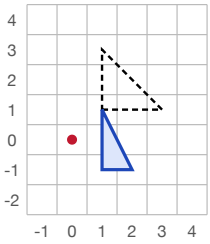
L6 · 2D transforms and frames

29. Compose 2D transformations and apply

29a Studying Graphics 2025, part 2 Q1

1. T(1, -2) = [1, 0, 1 ; 0, 1, -2 ; 0, 0, 1]
2. S(0.5, 1) = [0.5, 0, 0 ; 0, 1, 0 ; 0, 0, 1]
3. Composite (applied right-to-left): M = S(0.5, 1) · T(1, -2) = [0.5, 0, 0.5 ; 0, 1, -2 ; 0, 0, 1]
4. Column vectors $[x, y, 1]^T$: the FIRST transformation is the RIGHTMOST matrix. Order matters (T·R ≠ R·T).
5. M·$[1, 1, 1]^T$ = (0.5·1 + 0·1 + 0.5, 0·1 + 1·1 + -2) = (1, -1)
6. M·$[3, 1, 1]^T$ = (0.5·3 + 0·1 + 0.5, 0·3 + 1·1 + -2) = (2, -1)
7. M·$[1, 3, 1]^T$ = (0.5·1 + 0·3 + 0.5, 0·1 + 1·3 + -2) = (1, 1)

Answer: (1, -1), (2, -1), (1, 1)



29b Exam 1 2020 T4

1. T(1, -2) = [1, 0, 1 ; 0, 1, -2 ; 0, 0, 1]
2. R(45°) = [0.7071, -0.7071, 0 ; 0.7071, 0.7071, 0 ; 0, 0, 1]
3. Composite (applied right-to-left): M = R(45°) · T(1, -2) = [0.7071, -0.7071, 2.1213 ; 0.7071, 0.7071, -0.7071 ; 0, 0, 1]
4. Column vectors $[x, y, 1]^T$: the FIRST transformation is the RIGHTMOST matrix. Order matters (T·R ≠ R·T).
5. M·$[1, 1, 1]^T$ = (0.7071·1 + -0.7071·1 + 2.1213, 0.7071·1 + 0.7071·1 + -0.7071) = (2.1213, 0.7071)
6. M·$[3, 1, 1]^T$ = (0.7071·3 + -0.7071·1 + 2.1213, 0.7071·3 + 0.7071·1 + -0.7071) = (3.5355, 2.1213)
7. M·$[1, 3, 1]^T$ = (0.7071·1 + -0.7071·3 + 2.1213, 0.7071·1 + 0.7071·3 + -0.7071) = (0.7071, 2.1213)

Answer: (2.121, 0.707), (3.536, 2.121), (0.707, 2.121)

group	triangle, T	M (top two rows)	result
1	(1,1)(3,1)(1,3) T[1,−2]	[0.707 −0.707 2.121] [0.707 0.707 −0.707]	(2.121, 0.707) (3.536, 2.121) (0.707, 2.121)
2	(1,2)(3,2)(1,4) T[−1,2]	[0.707 −0.707 −2.121] [0.707 0.707 0.707]	(−2.828, 2.828) (−1.414, 4.243) (−4.243, 4.243)
3	(2,1)(4,1)(2,3) T[−1,−2]	[0.707 −0.707 0.707] [0.707 0.707 −2.121]	(1.414, 0) (2.828, 1.414) (0, 1.414)
4	(2,2)(4,2)(2,4) T[1,2]	[0.707 −0.707 −0.707] [0.707 0.707 2.121]	(−0.707, 4.95) (0.707, 6.364) (−2.121, 6.364)

29c EXAM 1/23 #3

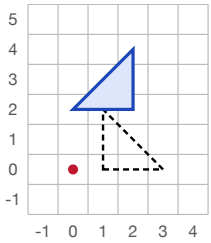
1. S(1.5, 0.5) about (-3, 2) = T(-3,2)·S(1.5, 0.5)·T(3,-2) = [1.5, 0, 1.5 ; 0, 0.5, 1 ; 0, 0, 1]
2. Composite (applied right-to-left): M = S(1.5, 0.5) = [1.5, 0, 1.5 ; 0, 0.5, 1 ; 0, 0, 1]
3. Column vectors [x, y, 1]^T: the FIRST transformation is the RIGHTMOST matrix. Order matters (T·R ≠ R·T).
4. M·[-5, 0, 1]^T = (1.5·-5 + 0·0 + 1.5, 0·-5 + 0.5·0 + 1) = (-6, 1)
5. M·[-1, 0, 1]^T = (1.5·-1 + 0·0 + 1.5, 0·-1 + 0.5·0 + 1) = (0, 1)
6. M·[-1, 4, 1]^T = (1.5·-1 + 0·4 + 1.5, 0·-1 + 0.5·4 + 1) = (0, 3)

Answer: (-6, 1), (0, 1), (0, 3)

29d practice (new numbers)

1. R(90°) = [0, -1, 0 ; 1, 0, 0 ; 0, 0, 1]
2. T(2, 1) = [1, 0, 2 ; 0, 1, 1 ; 0, 0, 1]
3. Composite (applied right-to-left): M = T(2, 1) · R(90°) = [0, -1, 2 ; 1, 0, 1 ; 0, 0, 1]
4. Column vectors [x, y, 1]^T: the FIRST transformation is the RIGHTMOST matrix. Order matters (T·R ≠ R·T).
5. M·[1, 0, 1]^T = (0·1 + -1·0 + 2, 1·1 + 0·0 + 1) = (2, 2)
6. M·[3, 0, 1]^T = (0·3 + -1·0 + 2, 1·3 + 0·0 + 1) = (2, 4)
7. M·[1, 2, 1]^T = (0·1 + -1·2 + 2, 1·1 + 0·2 + 1) = (0, 2)

Answer: (2, 2), (2, 4), (0, 2)



29e practice (new numbers)

1. S(2, 2) about (1, 1) = T(1,1)·S(2, 2)·T(-1,-1) = [2, 0, -1 ; 0, 2, -1 ; 0, 0, 1]
2. Composite (applied right-to-left): M = S(2, 2) = [2, 0, -1 ; 0, 2, -1 ; 0, 0, 1]
3. Column vectors [x, y, 1]^T: the FIRST transformation is the RIGHTMOST matrix. Order matters (T·R ≠ R·T).
4. M·[1, 1, 1]^T = (2·1 + 0·1 + -1, 0·1 + 2·1 + -1) = (1, 1)
5. M·[3, 1, 1]^T = (2·3 + 0·1 + -1, 0·3 + 2·1 + -1) = (5, 1)
6. M·[1, 3, 1]^T = (2·1 + 0·3 + -1, 0·1 + 2·3 + -1) = (1, 5)

Answer: (1, 1), (5, 1), (1, 5)

30. Change of coordinate frame

30a Studying Graphics 2025, part 2 Q2

1. Axes of the new frame (unit vectors, measured in OXY): u = [0, 1], v = [-1, 0] (frame rotated 90° CCW); origin E = (0, 0)
2. move to the new origin: P − E = [2, 2]
3. project on the axes: u = (P − E)·u = 2, v = (P − E)·v = -2
4. Matrix form: [u; v; 1] = [u_x u_y 0; v_x v_y 0; 0 0 1] · T(0, 0) · [x; y; 1] (rows = the new axes). Equivalently: rotate the POINT by −90°.

Answer: (u, v) = (2, -2)

> The 2025 guide multiplies R·P → (−2,2): that rotates the point, not the frame. The frame rotated +90° = the point rotated −90° → (2,−2). Check: the new u axis points along old +y, and P has y = 2 → u = 2 ✓.

30b practice (new numbers)

1. Axes of the new frame (unit vectors, measured in OXY): u = [0, -1], v = [1, 0] (frame rotated 90° CW); origin E = (0, 0)
2. move to the new origin: P − E = [3, 1]
3. project on the axes: u = (P − E)·u = -1, v = (P − E)·v = 3
4. Matrix form: [u; v; 1] = [u_x u_y 0; v_x v_y 0; 0 0 1] · T(0, 0) · [x; y; 1] (rows = the new axes). Equivalently: rotate the POINT by +90°.

Answer: (u, v) = (-1, 3)

30c practice (new numbers)

1. Axes of the new frame (unit vectors, measured in OXY): u = [0, 1], v = [-1, 0] (frame rotated 90° CCW); origin E = (1, 1)
2. move to the new origin: P − E = [3, 2]
3. project on the axes: u = (P − E)·u = 2, v = (P − E)·v = -3
4. Matrix form: [u; v; 1] = [u_x u_y 0; v_x v_y 0; 0 0 1] · T(-1, -1) · [x; y; 1] (rows = the new axes). Equivalently: rotate the POINT by −90°.

Answer: (u, v) = (2, -3)

L6 · Clipping

31. Cohen–Sutherland line clipping

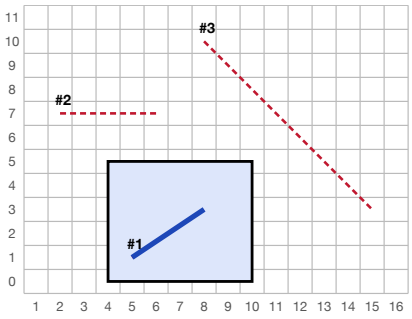
31a Studying Graphics 2025, part 2 Q3

Outcode bits [Top Bottom Right Left] (slide: 1001 = top-left, 0110 = bottom-right). Both 0000 → accept. AND ≠ 0000 → reject. Otherwise cut the outside end on one window edge (lecture order: left, top, right, bottom), recompute its code, repeat.

Segment 1: (5,1)–(8,3)
P1(5,1) code 0000, P2(8,3) code 0000; AND = 0000
both 0000 → ACCEPT (5,1)–(8,3)

Segment 2: (2,7)–(6,7)
P1(2,7) code 1001, P2(6,7) code 1000; AND = 1000
AND = 1000 ≠ 0000 → REJECT (whole segment outside on one side)

Segment 3: (8,10)–(15,3)
P1(8,10) code 1000, P2(15,3) code 0010; AND = 0000
AND = 0000 but not both 0000 → clip
clip P1 against top y = 5: $x = x_1 + (x_2–x_1)(y–y_1)/(y_2–y_1)$ → (13,5), new code 0010
new AND = 0010
AND = 0010 ≠ 0000 → REJECT (whole segment outside on one side)



31b Exam 1 2020 groups 1–4

group	segment 1	segment 2	segment 3
1	(3,1)–(13,3) → accept (4, 1.2)–(10, 2.4)	(2,8)–(6,8) → reject	(7,10)–(15,3) → reject
2	(3,2)–(13,4) → accept (4, 2.2)–(10, 3.4)	(2,6)–(6,6) → reject	(7,11)–(15,4) → reject
3	(2,3)–(12,4) → accept (4, 3.2)–(10, 3.8)	(1,6)–(7,6) → reject	(8,11)–(14,4) → reject
4	(3,2)–(11,4) → accept (4, 2.25)–(10, 3.75)	(3,7)–(7,7) → reject	(9,11)–(15,5) → reject

31c practice (new numbers)

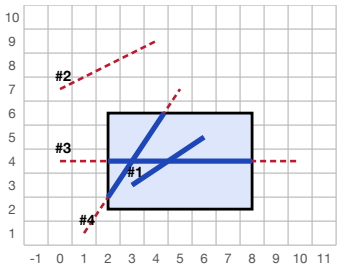
Segment 1: (3,3)–(6,5)
P1(3,3) code 0000, P2(6,5) code 0000; AND = 0000
both 0000 → ACCEPT (3,3)–(6,5)

Segment 2: (0,7)–(4,9)
P1(0,7) code 1001, P2(4,9) code 1000; AND = 1000
AND = 1000 ≠ 0000 → REJECT (whole segment outside on one side)

Segment 3: (0,4)–(10,4)
P1(0,4) code 0001, P2(10,4) code 0010; AND = 0000
AND = 0000 but not both 0000 → clip
clip P1 against left x = 2: $y = y_1 + (y_2–y_1)(x–x_1)/(x_2–x_1)$ → (2,4), new code 0000
new AND = 0000
clip P2 against right x = 8: $y = y_1 + (y_2–y_1)(x–x_1)/(x_2–x_1)$ → (8,4), new code 0000
new AND = 0000
both 0000 → ACCEPT (2,4)–(8,4)

Segment 4: (1,1)–(5,7)
P1(1,1) code 0101, P2(5,7) code 1000; AND = 0000
AND = 0000 but not both 0000 → clip
clip P1 against left x = 2: $x = x_1 + (x_2–x_1)(y–y_1)/(y_2–y_1)$ → (2,2.5), new code 0000
new AND = 0000

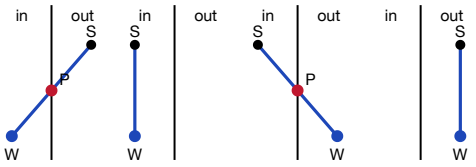
clip P2 against top y = 6: $x = x_1 + (x_2 - x_1)(y - y_1)/(y_2 - y_1) \rightarrow (4.3333, 6)$, new code 0000
new AND = 0000
both 0000 → ACCEPT (2,2.5)–(4.3333,6)



Answer: #1 kept (3,3)–(6,5) · #2 rejected · #3 kept (2,4)–(8,4) · #4 kept (2,2.5)–(4.33,6)

32. Sutherland–Hodgman polygon clipping

32a Studying Graphics 2025, part 2 Q4



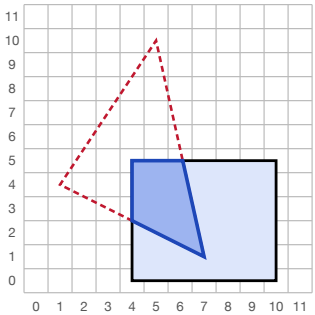
Clip against left x = 4
(5, 10)→(7, 1): in → in, output W (7, 1)
(7, 1)→(1, 4): in → out, output crossing P (4, 2.5)
(1, 4)→(5, 10): out → in, output P (4, 8.5) and W (5, 10)
→ (7, 1) (4, 2.5) (4, 8.5) (5, 10)

Clip against top y = 5
(5, 10)→(7, 1): out → in, output P (6.11, 5) and W (7, 1)
(7, 1)→(4, 2.5): in → in, output W (4, 2.5)
(4, 2.5)→(4, 8.5): in → out, output crossing P (4, 5)
(4, 8.5)→(5, 10): out → out, output nothing
→ (6.11, 5) (7, 1) (4, 2.5) (4, 5)

Clip against right x = 10
(4, 5)→(6.11, 5): in → in, output W (6.11, 5)
(6.11, 5)→(7, 1): in → in, output W (7, 1)
(7, 1)→(4, 2.5): in → in, output W (4, 2.5)
(4, 2.5)→(4, 5): in → in, output W (4, 5)
→ (6.11, 5) (7, 1) (4, 2.5) (4, 5)

Clip against bottom y = 0
(4, 5)→(6.11, 5): in → in, output W (6.11, 5)
(6.11, 5)→(7, 1): in → in, output W (7, 1)
(7, 1)→(4, 2.5): in → in, output W (4, 2.5)
(4, 2.5)→(4, 5): in → in, output W (4, 5)
→ (6.11, 5) (7, 1) (4, 2.5) (4, 5)

Answer: (6.11, 5), (7, 1), (4, 2.5), (4, 5) (55/9 = 6.11)



32b practice (new numbers)

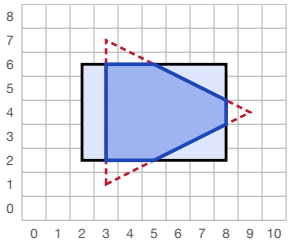
Clip against left x = 2
(3, 7)→(3, 1): in → in, output W (3, 1)
(3, 1)→(9, 4): in → in, output W (9, 4)
(9, 4)→(3, 7): in → in, output W (3, 7)
→ (3, 1) (9, 4) (3, 7)

Clip against top y = 6
(3, 7)→(3, 1): out → in, output P (3, 6) and W (3, 1)
(3, 1)→(9, 4): in → in, output W (9, 4)
(9, 4)→(3, 7): in → out, output crossing P (5, 6)
→ (3, 6) (3, 1) (9, 4) (5, 6)

Clip against right x = 8
(5, 6)→(3, 6): in → in, output W (3, 6)
(3, 6)→(3, 1): in → in, output W (3, 1)
(3, 1)→(9, 4): in → out, output crossing P (8, 3.5)
(9, 4)→(5, 6): out → in, output P (8, 4.5) and W (5, 6)
→ (3, 6) (3, 1) (8, 3.5) (8, 4.5) (5, 6)

Clip against bottom y = 2
(5, 6)→(3, 6): in → in, output W (3, 6)
(3, 6)→(3, 1): in → out, output crossing P (3, 2)
(3, 1)→(8, 3.5): out → in, output P (5, 2) and W (8, 3.5)
(8, 3.5)→(8, 4.5): in → in, output W (8, 4.5)
(8, 4.5)→(5, 6): in → in, output W (5, 6)
→ (3, 6) (3, 2) (5, 2) (8, 3.5) (8, 4.5) (5, 6)

Answer: (3, 6), (3, 2), (5, 2), (8, 3.5), (8, 4.5), (5, 6)



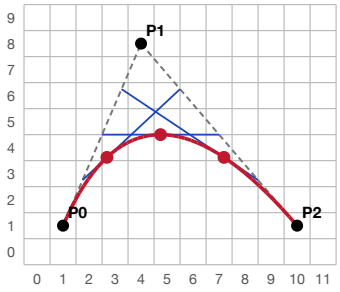
L6 · Bézier curves

33. Bézier curve construction (+ smooth joining)

33a Studying Graphics 2025, part 2 Q5

de Casteljau: on each control-polygon segment mark the point at fraction t, join those points, repeat until one point remains = B(t).

1. Degree 2 (quadratic = second order, 3 points). The curve starts at P0, ends at P2, and is pulled toward (but does not pass through) the inner control points.
2. Construction for a given t: on every segment of the control polygon mark the point at fraction t; join the new points and repeat until one point is left = B(t).
3. t = 0.25, round 1: (1.75, 2.75) (5.5, 6.25)
4. t = 0.25, round 2: (2.6875, 3.625)
5. t = 0.5, round 1: (2.5, 4.5) (7, 4.5)
6. t = 0.5, round 2: (4.75, 4.5)
7. t = 0.75, round 1: (3.25, 6.25) (8.5, 2.75)
8. t = 0.75, round 2: (7.1875, 3.625)
9. Formula: $B(t) = (1-t)^2P0 + 2t(1-t)P1 + t^2P2$
10. Smooth joining of a next curve Q: Q0 = P2 (10, 1) (C0, continuous), and Q1 on the line P1P2 beyond P2 (G1, same tangent direction). Equal length too (C1): Q1 = 2P2 − P1 = (16, -6).

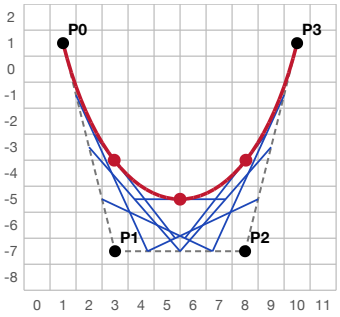


Order the points P_0 (1,1), P_1 (4,8), P_2 (10,1): the control point is the middle one (the 2025 guide puts (10,1) as P_1).

33b EXAM 1/22 #1

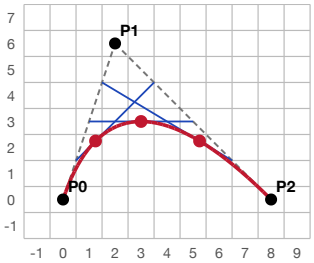
1. Degree 3 (cubic = third order, 4 points). The curve starts at P0, ends at P3, and is pulled toward (but does not pass through) the inner control points.
2. Construction for a given t: on every segment of the control polygon mark the point at fraction t; join the new points and repeat until one point is left = B(t).
3. t = 0.25, round 1: (1.5, -1) (4.25, -7) (8.5, -5)
4. t = 0.25, round 2: (2.1875, -2.5) (5.3125, -6.5)

5. t = 0.25, round 3: (2.9688, -3.5)
6. t = 0.5, round 1: (2, -3) (5.5, -7) (9, -3)
7. t = 0.5, round 2: (3.75, -5) (7.25, -5)
8. t = 0.5, round 3: (5.5, -5)
9. t = 0.75, round 1: (2.5, -5) (6.75, -7) (9.5, -1)
10. t = 0.75, round 2: (5.6875, -6.5) (8.8125, -2.5)
11. t = 0.75, round 3: (8.0313, -3.5)
12. Formula: $B(t) = (1-t)^3P0 + 3t(1-t)^2P1 + 3t^2(1-t)P2 + t^3P3$
13. Smooth joining of a next curve Q: Q0 = P3 (10, 1) (C0, continuous), and Q1 on the line P2P3 beyond P3 (G1, same tangent direction). Equal length too (C1): Q1 = 2P3 − P2 = (12, 9).



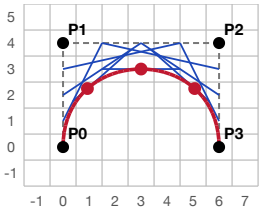
33c practice (new numbers)

1. Degree 2 (quadratic = second order, 3 points). The curve starts at P0, ends at P2, and is pulled toward (but does not pass through) the inner control points.
2. Construction for a given t: on every segment of the control polygon mark the point at fraction t; join the new points and repeat until one point is left = B(t).
3. t = 0.25, round 1: (0.5, 1.5) (3.5, 4.5)
4. t = 0.25, round 2: (1.25, 2.25)
5. t = 0.5, round 1: (1, 3) (5, 3)
6. t = 0.5, round 2: (3, 3)
7. t = 0.75, round 1: (1.5, 4.5) (6.5, 1.5)
8. t = 0.75, round 2: (5.25, 2.25)
9. Formula: $B(t) = (1-t)^2P0 + 2t(1-t)P1 + t^2P2$
10. Smooth joining of a next curve Q: Q0 = P2 (8, 0) (C0, continuous), and Q1 on the line P1P2 beyond P2 (G1, same tangent direction). Equal length too (C1): Q1 = 2P2 − P1 = (14, -6).



33d practice (new numbers)

1. Degree 3 (cubic = third order, 4 points). The curve starts at P0, ends at P3, and is pulled toward (but does not pass through) the inner control points.
2. Construction for a given t: on every segment of the control polygon mark the point at fraction t; join the new points and repeat until one point is left = B(t).
3. t = 0.25, round 1: (0, 1) (1.5, 4) (6, 3)
4. t = 0.25, round 2: (0.375, 1.75) (2.625, 3.75)
5. t = 0.25, round 3: (0.9375, 2.25)
6. t = 0.5, round 1: (0, 2) (3, 4) (6, 2)
7. t = 0.5, round 2: (1.5, 3) (4.5, 3)
8. t = 0.5, round 3: (3, 3)
9. t = 0.75, round 1: (0, 3) (4.5, 4) (6, 1)
10. t = 0.75, round 2: (3.375, 3.75) (5.625, 1.75)
11. t = 0.75, round 3: (5.0625, 2.25)
12. Formula: $B(t) = (1-t)^3P0 + 3t(1-t)^2P1 + 3t^2(1-t)P2 + t^3P3$
13. Smooth joining of a next curve Q: Q0 = P3 (6, 0) (C0, continuous), and Q1 on the line P2P3 beyond P3 (G1, same tangent direction). Equal length too (C1): Q1 = 2P3 − P2 = (6, -4).



Theory · true / false with a reason

34. Short theory / true-false

34a lecture tests, older papers

1. **FALSE**: a pixel is a sample (position + value); the display gives it a size.
2. **FALSE**: it only reduces it (finite raster).
3. **FALSE**: CMY is subtractive (ink); RGB is additive (light).
4. **TRUE**: a, b, p are integers; DDA needs floating point.
5. **TRUE**: $R(\alpha)R(\beta) = R(\alpha+\beta) = R(\beta)R(\alpha)$; but $T \cdot R \neq R \cdot T$ in general.
6. **FALSE**: AND = 0000 only means "cannot reject yet"; both codes 0000 means accept.
7. **FALSE**: only through P_0 and P_n; it stays inside the convex hull.
8. **FALSE**: raster redraws the whole frame; a vector display depends on the drawing.
9. **FALSE**: its gamut is the triangle of its primaries inside the horseshoe.
10. **FALSE**: that is the vertex shader; the fragment shader outputs a colour (RGBA, maybe depth).
11. **TRUE**: $[x\ y\ 1]^T$ with a 3×3 matrix.
12. **FALSE**: they lie on the curved edge; the straight "purple line" is not spectral.
13. **TRUE**: memory = W·H·bits/8.
14. **FALSE**: one window edge at a time; the output of one stage is the input of the next.